

What is MicroPulse?

A daily readiness and decision-support platform that tells you, every morning, who on your squad is ready to train, who needs managing, and what session each athlete should actually do today.

In one paragraph

MicroPulse pulls together everything you already collect about your athletes — wellness check-ins, GPS load from training and matches, force-plate data, RPE, sleep, and your medical notes — and converts it into a single GREEN / YELLOW / RED readiness state per player, every day. Coaches see who is good to go, who needs reduced load, and who should be on recovery work. The session each athlete sees on their phone is automatically scaled to that state. Match-day taper, week setup, and Return-to-Play progression are all driven by the same data, in the same dashboard.

5

Live integrations
Catapult, StatSports, VALD,
GymAware, Whoop

4

Scoring models
ACWR · MLI · Metabolic Load
· NM Fatigue

60 min

Setup to first decision
Contract → live

30 sec

Athlete daily input
Mobile check-in

Who uses it, and what they actually see

MicroPulse has four primary user roles. Each sees a different view of the same underlying data, sized to how much of their day this should occupy.

Role	What they see	Time spent / day
Athlete	30-second wellness check-in. Today's session, scaled to their state.	30–60 sec
Head coach	Squad-wide GREEN/YELLOW/RED list. Today's training plan.	2–3 min
S&C / performance	Per-player load, ACWR, force-plate trends. Custom programme builder.	10–20 min
Medical / physio	Return-to-play stage tracker. Wellness alerts. Audit log.	5–10 min

The next two pages walk through a typical Tuesday at a premier-league level club using MicroPulse, scene by scene, from each role's perspective.



1. A typical Tuesday (MD-3)

Three days before a Saturday match. Same data flows through all four roles — each sees just what they need.

ATHLETE — Magnús, midfielder

07:42, on the bus to training

```
How did you sleep?  
[ 1 2 3 4 5 6 7 ] Soreness  
[ 1 2 3 4 5 6 7 ] Fatigue  
Today: GREEN — Full session prescribed.  
→ MD-3 NEURAL / VELOCITY block  
Open session · RPE later
```

Magnús opens the MicroPulse app on the bus. Five swipes, thirty seconds. The system already knows his sleep duration from his Oura ring, his external load from yesterday's GPS, and his neuromuscular state from Friday's force-plate test. His check-in is the missing subjective layer. He gets a GREEN back — full session today — and his prescribed session appears on the screen, scaled to his state.

HEAD COACH — morning briefing

08:15, manager's office

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Today · Tuesday MD-3 · Squad readiness  
  
GREEN ..... 20 players  
YELLOW ... 3 players  
RED .. 2 players  
  
→ Andri H. YELLOW · ACWR drift, sleep -1.5h  
→ Birkir P. RED · RTP stage 3, contact watch
```

The head coach opens MicroPulse at his desk before staff meeting. One screen, one minute. Twenty players are GREEN: full session. Three are YELLOW (managed caution). Two are RED. He clicks Andri's name and sees the why — ACWR drifted to 1.6 on yesterday's session and he reported 1.5 hours less sleep than baseline. The recommendation is to cut his afternoon session by 25%. The coach approves with one tap.



S&C COACH — programme review

11:30, gym office

Custom templates · MD-3 NEURAL

A. Garcia-Ramos Cluster

15 × 1 · 85% 1RM · 6s rest

B. Potentiation Cluster

Trap-bar DL + Box Jump · 4×2

Auto-generated YELLOW variant:

→ Volume -30% · drop B block

The S&C coach designed the GREEN session for MD-3 last month — a Garcia-Ramos cluster followed by a potentiation block. MicroPulse has auto-generated the YELLOW variant (reduced volume, second block dropped) and RED variant (activation only). He reviews the auto-generated versions, tweaks the YELLOW to keep the cluster at 80% instead of 70%, saves. Done in five minutes. Andri will see the YELLOW version on his phone the moment he arrives at the gym.

MEDICAL — physio room

14:00, post-treatment

Return-to-Play · Birkir P. · Stage 3

Hamstring strain · day 18

Stage 0 → 1 → 2 → 3 → 4 → 5

Stage 3 criteria:

- ✓ No pain in straight-line running
- ✓ Wellness check-in ≥ 6/10 for 3 days
- × ForceDecks asymmetry < 8% (currently 11%)

The physio opens Birkir's RTP record. He's on day 18 of a hamstring rehab, currently at Stage 3 (non-contact drills). MicroPulse shows the criteria for moving to Stage 4 — two are met, but ForceDecks asymmetry is still 11% (threshold 8%). The physio decides to hold him at Stage 3, schedules another force-plate test for Thursday, and adds a note. Every action is timestamped and stored in an audit log that satisfies GDPR Article 30 record-keeping.

2. What MicroPulse does, end to end

Eight core capabilities. Each replaces something a typical club today does in a spreadsheet, a WhatsApp thread, or three different tools.

Capability	What it replaces	Used by
Daily readiness scoring	Coach gut feel; standalone wellness apps	Everyone
Adaptive session prescription	Manual “full / reduced / off” decisions per athlete	Coach, S&C
Custom training programme builder	Excel templates passed by email	S&C
Match-week setup (MD-N mapping)	Whiteboard plus mental arithmetic	Coach, S&C
6-stage Return-to-Play workflow	Word document; physio’s notebook	Medical
GPS / force-plate / wearable ingest	Manual export-import between tools	S&C, performance
Player-coach two-way messaging	WhatsApp groups	Everyone
Coach drill library (portable)	Coach’s personal Dropbox folder	Coach, S&C

All eight capabilities sit on top of the same underlying data model, so a wellness change is reflected in the readiness score, which is reflected in the prescribed session, which is reflected in the post-session RPE, which is reflected in tomorrow’s readiness. There is no manual data movement between tools.

Athlete delivery: PWA, no app-store install

The athlete experience is a Progressive Web App. The athlete opens the team URL once on their phone, taps “Add to Home Screen”, and the app behaves identically to a native app from then on — push notifications, offline check-ins, background sync. No App Store or Play Store approval, no version-update friction, no separate iOS / Android codebases. Squad onboarding takes about ten minutes per player, including data-protection consent.

3. How readiness is calculated

Four scoring models feed the daily GREEN / YELLOW / RED state. Each can be inspected on its own; none is a black box.

Acute-to-chronic workload ratio (ACWR)

Per Gabbett (2016) and Hulin et al. (2016). MicroPulse computes ACWR rolling per athlete with a 7-day acute window over a 28-day chronic window, EWMA-smoothed. Both classical and uncoupled variants are exposed in the dashboard. The system surfaces athletes drifting toward or above the 1.5 ceiling, and links the alert back to the specific session that pushed the ratio up — not just “you are at 1.6” but “Tuesday’s session added 480 AU of unbalanced load relative to your monthly base.”

Mechanical Load Index (MLI)

Quantifies the wear-and-tear component of GPS load: decelerations, accelerations, change-of-direction events, and PlayerLoad density per minute. MLI separates mechanical stress (joint, tendon, neural) from metabolic stress (energy systems) — because a player who runs 8 km steady-state has very different recovery needs from one who covers 6 km with 50 hard decelerations. Soft-tissue injury risk correlates more strongly with MLI than with total distance, which is why the model weights it heavily in the daily readiness score.

Metabolic Load Score

Estimates metabolic load via metabolic power (di Prampero, Osgnach et al.) which combines speed and acceleration into an instantaneous oxygen-demand proxy. High-Metabolic-Load (HML) distance — distance covered at a metabolic power of ~25 W/kg or higher — is then summed per session. The score also classifies the session as **metabolic-dominant** (steady-state distance, aerobic), **neural-dominant** (short, explosive, repeated sprints), or **mixed**. This matters because a metabolic-heavy day is followed by very different recovery work than a neural-heavy day, and the Adaptive Training Engine uses this classification to shape the next session.

Neural Fatigue Model

Force-plate-driven freshness signal from VALD ForceDecks countermovement-jump testing. Tracks jump height, contraction time, reactive strength index ($RSI = \text{jump height} / \text{contact time}$), and left-right asymmetry. Trends are computed against the athlete’s personal baseline rather than population norms, so a small but consistent dip is flagged before it would show in subjective wellness. In practice the neural signal often leads subjective fatigue by 24–48 hours, which is the window where preventive load adjustment is most valuable.

Adaptive Training Engine

The orchestration layer that turns the four scores above into a session prescription. The coach designs the GREEN session for each match-day slot once. For each athlete each morning, the engine reads their current readiness state and the dominant signal driving it, then generates the YELLOW and RED variants accordingly: a wellness-driven YELLOW reduces volume; an ACWR-driven YELLOW reduces intensity; a NM-fatigue-driven YELLOW drops maximal blocks while keeping technical work; a RED state regresses to recovery work appropriate to the cause (active recovery for metabolic dominance, mobility for mechanical dominance). Coaches can override any auto-generated session in one click.

4. Integration coverage

MicroPulse is not a wearable. It reads from the providers your club already uses, normalises the data, and stores it under your tenant.

Provider	Category	Status
Catapult	GPS + external load (outdoor)	Live
StatSports	GPS + external load (outdoor)	Live
Polar Team Pro	GPS + heart rate (outdoor)	Foundation built; partner credentials in process
VALD ForceDecks	Force-plate / CMJ	Live
VALD ForceFrame	Isometric strength	Live
GymAware	Velocity-based training	Live
Whoop	Recovery, sleep, HRV	Live
Hudl Sportscore	Match-tagged events	On roadmap (Q3 2026)

5. Return-to-Play workflow

Six clinical stages with explicit per-stage progression criteria, adapted from current consensus. Each stage transition is timestamped and stored in an audit trail — useful for internal review, federation reporting, and insurance.

Stage	Label	Criteria
0	Rest	No physical activity
1	Light aerobic	Walking, swimming, cycling — no resistance
2	Sport-specific	Linear running, straight-line movement
3	Non-contact drills	Technical work, team drills without contact
4	Full training	Contact training, simulated play
5	Match play	Cleared, full participant

6. Technical architecture and data ownership

- **EU-only data residency.** All data stored on Supabase / PostgreSQL in Frankfurt. No data leaves the EEA.
- **Multi-tenant isolation at the database engine level** via PostgreSQL row-level security — each club's data is isolated in the database itself, not at the application layer.
- **Athlete owns their data.** Five granular consent types managed by the athlete in real time through the in-app Privacy panel. Append-only audit log per GDPR Article 30. Database-level trigger blocks data access for athletes under 18 without a parent/guardian consent record.
- **Controller / processor split.** The club is the data controller. Helgasport ehf. (operating MicroPulse) is the data processor. Standard DPA is provided alongside the service contract.
- **Coach drill ownership.** The drill library a coach builds in MicroPulse follows them between clubs. RLS is keyed on coach_id, not team_id.

7. Honest scope

What MicroPulse is and is not. Useful before the first call.

MicroPulse is:

- A daily readiness and decision-support layer on top of the monitoring tools you already use
- An RTP workflow tool that documents stage progression and creates a defensible audit trail
- A coach-led platform: the practitioners who use it shape it, not a separate analytics team
- A vehicle for keeping athlete consent, data ownership, and EU residency at the centre of how you handle player data

MicroPulse is not:

- A tactical analysis or video-coding platform — we integrate with Hudl and Sportscore rather than replace them
- A medical record system — it complements, but does not replace, your existing EMR
- A wearable manufacturer — we read data from existing providers, we do not sell sensors
- A black box — every readiness decision links back to the specific signals that drove it

8. Commercial model and roadmap

Subscription-based, billed per team. Three tiers (Free, Pro, Elite) with unlimited squad sizes on paid plans. A founder cohort with a 24-month price-lock is currently open for pre-launch reference customers — we can discuss commercial details in a follow-up tailored to your operating context.

Q2–3 2026 roadmap: Polar Team Pro live (partner credentials in flight), Hudl Sportscore XML import for automated match-minute and event ingestion, and federation-level deployments with national governing bodies. Beta access is offered to early customers ahead of general availability.

9. Next steps

We propose a 30-minute video walk-through of MicroPulse against the workflow of your medical and performance staff. If the fit is there, we offer a structured 8-week pilot at no cost, with weekly check-ins and a written results report at the end. The pilot can run alongside your existing AMS without displacing it. To schedule a walk-through, contact helgi@micropulse.is or visit micropulse.is.

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